

SECTION B

7 (**short**) A pendulum shock testing machine consists of a mass m on the end of a light rigid rod of length l that swings freely to impact a viscous buffer of damping rate λ located directly below the pendulum pivot, as shown in Fig. 4.

- (a) Calculate the velocity of the mass at the moment of impact with the buffer if the pendulum is released from rest at the horizontal position. [1]
- (b) Derive the equation of motion of the mass in terms of its velocity after the impact and calculate the peak acceleration. [5]
- (c) Stating your assumptions, calculate the distance travelled by the mass. [4]

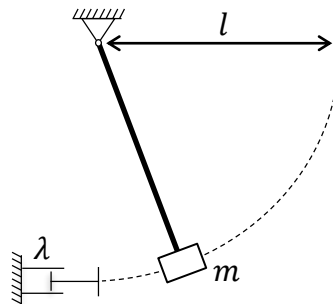


Fig. 4